

Couplings in Distributive Copy-Discard Categories

Elena Di Lavoie

February 19, 2026

We define couplings of morphisms in a distributive copy-discard category and prove that they form a category. Our definition of coupling is a general version of the one that appears in the program logics literature for the case of subdistributions.

Couplings. A distributive monoidal category always has another monoidal product defined by $A * B := A \otimes B + A + B$. If the original monoidal product, $\otimes$, has discard morphisms and the initial object is also terminal, then there are morphisms, $\pi_A^*: A * B \rightarrow A$ and $\pi_B^*: A * B \rightarrow B$, that resemble projections.

$$\pi_A^* := [\pi_A, \pi_A^+] = \begin{array}{c} AB \\ A \\ B \end{array} \begin{array}{c} \boxed{\pi_A} \\ \circ \\ \circ \end{array} \begin{array}{c} \circ \\ \circ \\ \circ \end{array} \begin{array}{c} A \\ \\ \end{array}$$

$$\pi_B^* := [\pi_B, \pi_B^+] = \begin{array}{c} AB \\ A \\ B \end{array} \begin{array}{c} \boxed{\pi_B} \\ \circ \\ \circ \end{array} \begin{array}{c} \circ \\ \circ \\ \circ \end{array} \begin{array}{c} B \\ \\ \end{array}$$

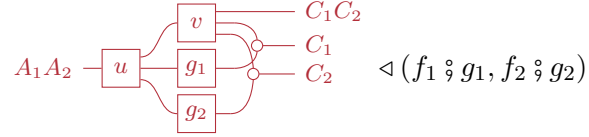
The morphisms $\pi_A: A \otimes B \rightarrow A$ and $\pi_B: A \otimes B \rightarrow B$ are the $\otimes$ -projections given by the discard morphisms and the morphisms $\pi_A^+: A + B \rightarrow A$ and $\pi_B^+: A + B \rightarrow B$ are the $+$ -projections given by the zero object.

A coupling of two morphisms, f_1 and f_2 , is a morphism that marginalises to f_1 on one side and to f_2 on the other. If we computed marginals using the $\otimes$ -projections, we would impose that f_1 and f_2 have the same failure behaviour. This is sometimes undesirable, so we use the $*$ -projections.

Definition 0.1. Consider two morphisms, $f_1: A_1 \rightarrow B_1$ and $f_2: A_2 \rightarrow B_2$, in a distributive copy-discard category $\mathbb{C}$ with zero. A *coupling* of (f_1, f_2) is a morphism $u: A_1 \otimes A_2 \rightarrow B_1 \otimes B_2 + B_1 + B_2$ such that $u \circ \pi_1^* = \pi_1 \circ f_1$ and $u \circ \pi_2^* = \pi_2 \circ f_2$. In this case, we write $u \triangleleft (f_1, f_2)$.

A category of couplings. We start by defining compositions and identities of couplings.

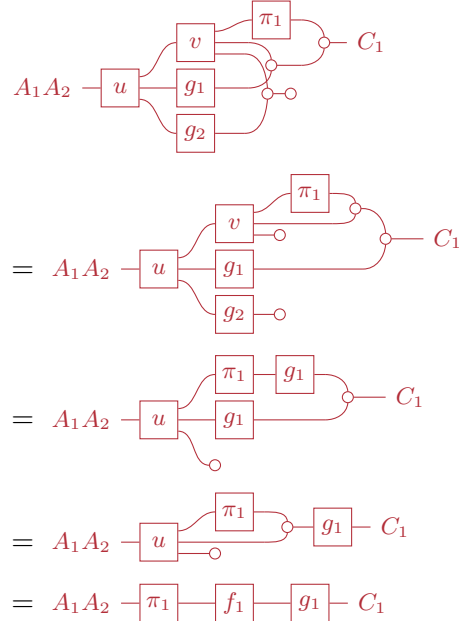
Lemma 0.2 (Coupling composition). *In a distributive copy-discard category with zero, couplings compose: for two couplings, $u \triangleleft (f_1, f_2)$ and $v \triangleleft (g_1, g_2)$, of morphisms $f_i: A_i \rightarrow B_i$ and $g_i: B_i \rightarrow C_i$, there is a coupling of the compositions, $u \circ [v, \text{id} + g_1 + g_2] \triangleleft (f_1 \circ g_1, f_2 \circ g_2)$.*



Proof. We use that copairings are associative and natural, that the initial object is also terminal, and that u and v are couplings.

$$\begin{aligned}
& u \circ [v, \text{id} + g_1 + g_2] \circ \pi_1^* \\
&= u \circ [v, \text{id} + g_1 + g_2] \circ [\pi_1, \pi_1^+] \\
&= u \circ [v \circ \pi_1^*, (g_1 + g_2) \circ \pi_1^+] \\
&= u \circ [\pi_1 \circ g_1, \pi_1^+ \circ g_1] \\
&= u \circ [\pi_1, \pi_1^+] \circ g_1 \\
&= u \circ \pi_1^* \circ g_1 \\
&= \pi_1 \circ f_1 \circ g_1
\end{aligned}$$

We may also use string diagrams for the coproduct.



The proof of the other side is similar. □

Lemma 0.3 (Identity coupling). *In a distributive copy-discard category with zero, there is always a coupling of two identities, $\iota_{A \otimes B} \triangleleft (\mathbb{1}_A, \mathbb{1}_B)$.*

$$\iota_{A \otimes B} = \begin{array}{c} AB \text{ --- } AB \\ \text{--- } \bigcirc \text{--- } A \\ \text{--- } \bigcirc \text{--- } B \end{array}$$

Proof. With string diagrams, it is easy to see that

$$\iota_{A \otimes B} \circ \pi_A^* = \begin{array}{c} AB \text{ --- } \boxed{\pi_A} \\ \text{--- } \bigcirc \text{--- } A \\ \text{--- } \bigcirc \text{--- } \end{array} = \pi_A$$

and that

$$\iota_{A \otimes B} \circ \pi_B^* = \begin{array}{c} AB \text{ --- } \boxed{\pi_B} \\ \text{--- } \bigcirc \text{--- } B \\ \text{--- } \bigcirc \text{--- } \end{array} = \pi_B.$$

□

Proposition 0.4 (Category of couplings). *For a distributive copy-discard category with zero, $\mathbb{C}$, there is a category, $\mathbf{Couple}_{\mathbb{C}}$, where objects are pairs, (A_1, A_2) , of objects of $\mathbb{C}$ and morphisms are triples, $(u, f_1, f_2): (A_1, A_2) \rightarrow (B_1, B_2)$, of two morphisms in $\mathbb{C}$, $f_1: A_1 \rightarrow B_1$ and $f_2: A_2 \rightarrow B_2$, and a coupling of them, $u \triangleleft (f_1, f_2)$.*

Proof. We check that the compositions and identities defined in Lemmas 0.2 and 0.3 are associative and unital. This is easy to see with string diagrams. For unitality, one uses that the maps from the initial object are natural.

$$\begin{aligned} & (u, f_1, f_2) \circ \mathbb{1}_{(B_1, B_2)} \\ &= \left(\begin{array}{c} A_1 A_2 \text{ --- } \boxed{u} \\ \text{--- } \bigcirc \text{--- } B_1 B_2 \\ \text{--- } \bigcirc \text{--- } B_1 \\ \text{--- } \bigcirc \text{--- } B_2 \end{array}, f_1 \circ \mathbb{1}_{B_1}, f_2 \circ \mathbb{1}_{B_2} \right) \\ &= (u, f_1, f_2) \end{aligned}$$

$$\begin{aligned} & \mathbb{1}_{(A_1, A_2)} \circ (u, f_1, f_2) \\ &= \left(\begin{array}{c} A_1 A_2 \text{ --- } \boxed{u} \\ \text{--- } \bigcirc \text{--- } B_1 B_2 \\ \text{--- } \bigcirc \text{--- } B_1 \\ \text{--- } \bigcirc \text{--- } B_2 \end{array}, \mathbb{1}_{A_1} \circ f_1, \mathbb{1}_{A_2} \circ f_2 \right) \\ &= \left(\begin{array}{c} A_1 A_2 \text{ --- } \boxed{u} \\ \text{--- } \bigcirc \text{--- } B_1 B_2 \\ \text{--- } \bigcirc \text{--- } B_1 \\ \text{--- } \bigcirc \text{--- } B_2 \end{array}, f_1, f_2 \right) \end{aligned}$$

$$= (u, f_1, f_2)$$

For associativity, one uses that the copairings are natural and associative.

$$\begin{aligned}
& ((u, f_1, f_2) \circ (v, g_1, g_2)) \circ (w, h_1, h_2) \\
&= \left(\begin{array}{c} \text{Diagram 1} \end{array}, (f_1 \circ g_1) \circ h_1, (f_2 \circ g_2) \circ h_2 \right) \\
&= \left(\begin{array}{c} \text{Diagram 2} \end{array}, f_1 \circ g_1 \circ h_1, f_2 \circ g_2 \circ h_2 \right) \\
&= \left(\begin{array}{c} \text{Diagram 3} \end{array}, f_1 \circ (g_1 \circ h_1), f_2 \circ (g_2 \circ h_2) \right) \\
&= (u, f_1, f_2) \circ ((v, g_1, g_2) \circ (w, h_1, h_2))
\end{aligned}$$

□

Future work. We would like to look at monoidal structures of couplings. It seems that they have a $\otimes$ -monoidal structure, but not a $+$ -monoidal structure in an obvious way.